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To cite this article: Julie Rozenberg, Stéphane Hallegatte, Baptiste Perrissin-Fabert & Jean-Charles Hourcade (2013) Funding low-carbon investments in the absence of a carbon tax, Climate Policy, 13:1, 134-141, DOI: [10.1080/14693062.2012.691222](https://doi.org/10.1080/14693062.2012.691222)

To link to this article: <https://doi.org/10.1080/14693062.2012.691222>



Published online: 08 Jun 2012.



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Funding low-carbon investments in the absence of a carbon tax

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Introducing a carbon tax is difficult, partly because it suggests that current generations have to make sacrifices for the benefit of future generations. However, the climate change externality could be corrected without such a sacrifice. It is possible to set a carbon value, and use it to create 'carbon certificates' that can be accepted as part of commercial banks' legal reserves. These certificates can be distributed to low-carbon projects, and be exchanged by investors against concessional loans, reducing capital costs for low-carbon projects. As the issuance of carbon certificates would increase the quantity of money, it will either lead to accelerated inflation or induce the Central Bank to raise interest rates. Low-carbon projects will thus have access to cheaper loans at the expense of either 'regular' investors (in case of higher interest rates) or of lenders and depositors (in case of accelerated inflation). Within this scheme, mitigation expenditures are compensated by a reduction in regular investments, so that immediate consumption is maintained. It uses future generation wealth to pay for a hedge against climate change. This framework is not as efficient as a carbon tax but is politically easier to implement and represents an interesting step in the trajectory towards a low-carbon economy.

Keywords: abatement strategies; carbon finance; climate change; intergenerational equity; investment; policy frameworks; public policy

La mise en place d'une taxe carbone est difficile, en partie parce qu'elle suggère que les générations actuelles doivent faire des sacrifices au profit des générations futures. Il est cependant possible de corriger l'externalité du changement climatique sans faire de tels sacrifices, en établissant une valeur du carbone et en l'appliquant à des « certificats carbone » qui seraient acceptés comme réserves légales des banques commerciales. Ces certificats seraient distribués aux projets bas-carbone et échangés par les investisseurs contre des prêts concessionnels auprès de banques commerciales, réduisant ainsi le coût des projets. Puisque la distribution des certificats carbone augmenterait la quantité de monnaie en circulation, cela provoquerait une hausse de l'inflation ou forcerait la Banque Centrale à augmenter les taux d'intérêt. Les projets bas-carbone auraient ainsi accès à des prêts moins chers, au détriment des autres investisseurs (dans le cas d'une augmentation des taux d'intérêt) ou des épargnants (dans le cas d'une hausse de l'inflation). Dans ce cadre, les coûts de l'atténuation sont compensés par une diminution des investissements ordinaires, de façon à ce que la consommation soit maintenue. La richesse future est ainsi utilisée pour s'assurer contre le changement climatique. Ce mécanisme n'est pas aussi efficace qu'une taxe carbone mais est plus acceptable politiquement et représente un premier pas intéressant vers une économie bas-carbone.

Mots clés: Stratégies d'abattement; finance carbone; changement climatique; équité intergénérationnelle; investissement; cadres de politiques; politique publique

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1. Introduction

Climate change is now widely recognized as a threat to the environment, economic growth, and social welfare. In response, many countries have set individual long-term objectives or commitments in terms of future GHG emissions. The EU has made a commitment to reduce its emissions by 20% in 2020 relative to 1990 levels.¹ France and the UK have a long-term objective to reduce their emissions by a factor of four by 2050. Collectively, the countries of the world have decided to aim to limit global warming to 2 °C above the pre-industrial global temperature. To achieve this, world emissions need to be halved by 2050 (O'Neill et al., 2010). These targets are very ambitious and require immediate and energetic political action.

Reaching these targets will require significant investments, in particular in energy, building, transport, and end-use equipment. The World Development Report 2010 (World Bank, 2010) estimates that the incremental costs of mitigation in developing countries could reach US\$140–175 billion a year by 2030. When added to current needs, these costs will lead to total financing requirements of about \$264–563 billion over the same period. However, these numbers pale in significance compared to the current global Gross Fixed Capital Formation, which amounts to about \$11 trillion per year, suggesting that the challenge is not so much to increase global investments but rather to shift them towards low-carbon projects.² Such an action would be facilitated greatly by attributing a price to carbon emissions. Indeed, the High-Level Advisory Group on Climate Change Financing (AGF, 2010) emphasized the importance of a carbon price in the range of \$20–25/tCO₂e in 2020 in order to succeed in leveraging \$100 billion per year dedicated to climate finance.

As well as its impact on low-carbon investment,³ attributing a price to carbon emissions would help to coordinate the ambitious transition towards a low-carbon economy. This price could also influence household behaviour, change optimal production processes, and help local authorities coordinate their action. Because climate change is an environmental externality, it can be solved theoretically in an optimal way by internalizing the externality with a carbon price (Nordhaus, 1994; Rezai et al., 2009). Moreover, because climate change is not the only externality in the economic system (Hallegatte et al.,

2011), additional measures are needed to deal with other market failures, such as an R&D subsidy to cope with knowledge spillover.

Initially, it might appear attractive to create this carbon price through the introduction of a carbon tax. The advantages of this include the universal applicability, simplicity, efficacy, and low set-up costs of the carbon tax by virtue of existing administrative institutions. However, the carbon tax suffers from a lack of political acceptability, as illustrated by the failure to introduce it in France in 2010. Two main difficulties have to be overcome. The first relates to intragenerational equity, as a carbon tax may have negative redistributive effects between contemporary businesses and households. Indeed, correcting the externality can be done in such a way that it maximizes social welfare (aggregated over all individuals and discounted over time), although it may decrease the welfare of some individuals (in other terms, it is not Pareto improving).⁴ Zero-cost lump-sum interindividual transfers are required to make a welfare-maximizing solution Pareto-improving (e.g. Harberger, 1978). In spite of technical difficulties (Kanbur, 2010), past experience in relation to fossil-fuel subsidy removal has shown that these distributional effects can be compensated through *ad hoc* measures or pre-existing safety nets (Hope and Singh, 1995).

The second difficulty arises from intergenerational distributive impacts, which are more complicated to tackle. The perceived necessity to pay now for an expensive transition in exchange for remote future benefits is one of the obstacles to the implementation of ambitious climate policies. In a period of economic crisis, this trade-off may appear particularly unappealing to voters and decision makers.

As a consequence of this political unacceptability – which can be understood as a government failure – a carbon tax⁵ appears unrealistic in the current context, and other tools may have to be used to address the climate change issue, at least in the short term. In particular, several initiatives have been developed to shift global (public and private) investments towards low-carbon projects in the absence of a carbon price. Even though this is a second-best option, it represents significant progress compared to a business-as-usual scenario.

Private initiatives are essential for seeking out and implementing least-cost climate mitigation options.

However, they require public policies to establish the incentive frameworks that are necessary to catalyse high levels of private investment. Today's low-carbon projects suffer from a lack of funding because of market failures affecting innovation and dissemination of new technologies, high-risk perceptions resulting from uncertainties about future climate policies and carbon prices, and excess upfront costs that make them unable to offer returns that are commensurate with their (perceived) risk. The challenge is to convince investors to change the technical content of their projects by providing them with cheaper capital, for example by using limited public resources as leverage for private capital. A paper by the World Bank, IMF and OECD (2011) lists a number of innovative instruments that have been proposed to achieve this. For example, subsidies on fossil-fuel use can be redirected to public climate finance, market-based instruments for international aviation and maritime bunker fuels can be used as an innovative source of climate finance, carbon offset markets can play an important role, and there are significant opportunities for Multilateral Development Banks to mobilize resources through new pooled financing arrangements (e.g. Climate Investments Funds and the Global Environment Fund).

This article proposes another approach to provide cheaper capital to low-carbon projects – the use of differentiated interest rates. The appeal of this solution is that it can be designed in a way that avoids intergenerational distributive impacts and is thus more politically acceptable than other options. Because GHG emissions and climate change are an uncorrected negative externality, climate change mitigation can be Pareto-improving – with all generations benefiting from action – provided that intergenerational transfers are implemented (Sinn, 2007; Rezai et al., 2009). Our proposal is based on the idea that such transfers are possible through manipulation of the interest rate. Although it comes at the expense of aggregate efficiency (efficiency being measured using social welfare), this option is easier to implement because it is Pareto-improving⁶ compared with a business-as-usual scenario.

In practice, the carbon externality can be corrected by mitigation expenditures and investments, at the cost of a reduced total economic output over the short term. The intergenerational distributional impacts of mitigation actions can then be prevented, and the welfare cost for

current generations cancelled, if total investments are reduced such that immediate consumption is not affected.⁷ On the one hand, such an action would impact future generations negatively by providing them with less capital, but on the other hand, it would bring them the benefit of lower climate change impacts, while protecting the consumption of the current generation.⁸

In order to maintain current consumption, it is possible to manipulate interest rates so that capital becomes more expensive for carbon-emitting investments, and consumption is encouraged. Low-carbon investments could also benefit from lower interest rates (i.e. lower capital cost), providing an incentive to reduce GHG emissions.⁹ From a welfare maximization perspective, this solution is suboptimal because the carbon externality is internalized only in the cost of capital, and not in all relevant decisions. Also, the impact on discounted social welfare of these intergenerational transfers is ambiguous (e.g. this impact depends on the discount rate and on the form of the utility function). This option nevertheless has the potential to bring the economy closer to its social-welfare frontier, reducing the welfare consequences of the carbon externality.¹⁰

This article proposes an institutional framework in which low-carbon projects have access to cheaper loans. This framework can be set up in a single country, at the European level, or even at the international level.¹¹

2. Carbon certificates as a tool to finance low-carbon projects

The first step would be to set a carbon value. However, this would not be applied directly to all prices in the economy through a carbon tax, but would be used to value new assets that we call *carbon certificates*. These carbon certificates would have a fixed face value and would be considered as legal reserve assets that can be used by commercial banks to respect legal-reserve regulatory constraints. They would be created and allocated by an independent monitoring unit to low-carbon project investors at the beginning of their project, with the number of certificates depending on the project contribution to carbon emissions mitigation. This poses the problem of determining a metric to monitor, control, and verify projects, which we address briefly in Example 1. The independent

Example 1 The need for MRV standards

Measuring the contribution of a project to GHG emission reduction is extremely difficult, as suggested by the Clean Development Mechanism (CDM) experience (Schneider, 2009). Accurate project-based accounting of avoided emissions on a case-by-case basis is hardly feasible and would be very costly to perform. Avoided emissions are necessarily calculated from a controversial baseline, and indirect effects of investments cannot be fully integrated in projects' appraisals (Shalizi and Lecocq, 2009). Using avoided carbon emissions as bank reserves requires a very high standard of measurement and verification that has not been achieved with CDMs.

A more robust alternative approach is to define a taxonomy of low-carbon projects and conventionally attribute a number of carbon certificates to each type of project, regardless of whether the project is 'additional' or not. For instance, a photovoltaic electric plan would be allocated a number of certificates as a function of its production capacity and the average carbon content of electricity production in the country. In the same way, an urban public transportation investment (e.g. a new metro line) would receive certificates as a function of the expected number of passengers, its electricity consumption, and the average carbon content of electricity production in the country. Using such a typology of projects would make it unnecessary to carry out a detailed analysis of each project. It would allow the simplification of procedures, reduce transaction costs and project uncertainty, and mitigate fraud risks. However, the entire scheme is dependent on a well-defined typology and on appropriate rules and procedures. In particular, the procedures need to be able to adjust to new information and technical change (e.g. accommodate a new way of producing renewable energy).

monitoring unit would also fix a lifetime to the certificates, as a function of the project category.

As depicted in Figure 1, an investor who has been allocated carbon certificates can give them to a bank in exchange for a concessional loan. The bank would provide such a loan because it can then use the certificates as a reserve asset to reduce its capital costs and respect legal-reserve regulatory constraints.

Legal reserve requirements are an instrument in the hands of Central Banks to control money supply, in addition to their open-market operations that target a given short-term interest rate (Bank of England, 2011; European Central Bank, 2011; Federal Reserve Bank of

San Francisco, 2004). The minimum reserves each commercial bank must hold at the Central Bank are set as a counterpart to customer deposits and notes. If r equals the required reserves–deposits ratio, e the excess reserves–deposits ratio, and c the currency–deposits ratio, then the amount of money $M1$ (currency plus chequable deposits) is proportional to MB (monetary base) through the money multiplier m (Mishkin, 2010):

$$M1 = m MB \quad \text{with } m = \frac{1 + c}{r + e + c} \quad (1)$$

If reserves MB are exogenously increased by carbon

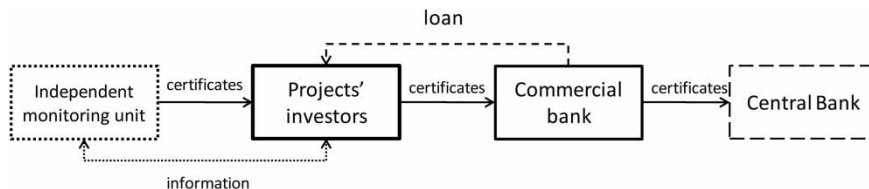


FIGURE 1 An investor receiving carbon certificates can exchange them for a concessional loan. Banks can use the certificates as legal reserve assets

certificates, and if the Central Bank maintains its target short-term interest rate ($\bar{i}$), then banks will expand deposits $M1$ through additional loans unless bank customers withdraw currency (i.e. an increase in c), or banks will decide to keep these reserves as excess reserves (i.e. an increase in e). If c , e , r , and $\bar{i}$ remain unchanged, then an increase in MB by $x\%$ translates into an increase in $M1$ by $x\%$. Attributing carbon certificates to low-carbon projects would thus increase MB , and in turn increase $M1$ (if $\Delta MB = C$, where C is carbon certificates in the Central Bank, then $(M1 + \Delta M1) = m(MB + \Delta MB)$ and $\Delta M1 = mC$). Because legal reserves influence demand for MB (even if r remains unchanged), they are one propagation channel for monetary policy that targets a given short-term interest rate (i.e. the price of MB).

We suggest that using legal reserves would be an efficient tool with which to finance mitigation. For a commercial bank, receiving carbon certificates from a project investor allows legal reserves and the amount of loans to increase, thereby increasing revenues (Example 2 gives an example of the potential gains for both the investor and the commercial bank). This system therefore provides a strong incentive for commercial banks to fund low-carbon projects with lower-rate loans. Accordingly, the lower the reserve ratio r , the more additional loans can be made for each carbon certificate. When r is very low, or in the most extreme case when $r = 0$, as in the UK, banks are not constrained by reserve requirements; however, loans are limited by the Basel III ratios (for instance, the capital ratio sets the maximum risk-weighted assets a bank can hold as a counterpart of its capital). In this case, carbon certificates just provide the bank with free liquidities, but the Basel III ratios might deteriorate if

low-carbon investments have a higher risk than regular investments.

Carbon certificates are not redeemable – they cannot be sold to the Central Bank and their face value stems only from their status as legal reserve. However, carbon certificates allow banks to increase their revenues, so a use value can be attributed to them. If a project was to be financed by savings or equity only, the investor would be allowed to sell certificates to a commercial bank, or to another low-carbon investor who needs to borrow from a commercial bank. This mechanism would provide financial support to low-carbon projects, even when no loan is needed. Note that there is no reason why the use value of certificates should be equal to their face value (i.e. the legal carbon price, which determines the value of carbon certificates as legal reserves assets). The selling price of certificates among investors would reflect the value they attach to an access to concessional loans, or the value of additional reserves for commercial banks.

The distribution of carbon certificates will create money to finance low-carbon investment. The quantity of additional money is determined by the legal carbon value and the number of available projects. To control this money creation, a limit on the number of certificates could be introduced, or the number of certificates conventionally attributed to various projects could be adjusted. Accordingly, this framework can only be instituted in a country where the Central Bank is able to implement efficient monetary policies. Also, confidence in monetary policy must be high enough to eliminate expectations of runaway inflation.

Of course, this mechanism is not magic and does not make low-carbon investments possible for free. There is

Example 2 Example of a transaction with carbon certificates

Suppose a low-carbon project is entitled to \$100 in certificates (e.g. $\$20/tCO_2 \times 5tCO_2$) and needs an investment of \$4000. The investor gives the \$100 of certificates to a commercial bank, which can gradually put them in its Central Bank account as the project progresses to completion. If the reserve ratio (r) is 10%, then the commercial bank can issue \$1000 'for free' and only needs to take \$1000 from other loans (thus reducing its opportunity cost). Let i be the interest rate. A 'surplus', equal to $i \times 1000$, has to be shared between the bank and the investor: a part of the \$1000 will be lent for free to the investor, thus reducing the loan real interest rate i_r , and the rest will be lent by the bank to other projects at the interest rate i . If $i_r = 3000/4000 \times i$ (i.e. the bank lends \$1000 for free to the investor) then all the surplus goes to the investor. By contrast, if $i_r = i$ the whole surplus goes to the bank. The loan real interest rate will then be defined as $0.75i < i_r < i$.

an additional cost to these projects, relative to the projects that would be implemented in the absence of GHG constraints, and this additional cost can be felt through accelerated inflation due to increased money creation. Whether or not the Central Bank reacts to prevent or counter this additional inflation will determine who pays for the mitigation policy. Note that this scheme would not modify monetary policies or central banking procedures (or ask them to take into account environmental issues), so it is unlikely to meet opposition from central bankers.

If the Central Bank increases its inflation target and leaves the nominal short-term interest rate constant, an 'inflation tax' will finance low-carbon projects at the expense of lenders and depositors. The scenario of an increased inflation target has been suggested by Blanchard et al. (2010), who question the low inflation policy that has been adopted by Central Banks for over a decade and posit that a 4% inflation rate might not be more costly than a 2% rate. Moreover, in Europe, this additional inflation could be seen as a way out of the 2011/2012 debt crisis, and generate more activity in the current situation of underemployment and underutilization of production capacities.¹²

If increased inflation is seen as unacceptable or undesirable, additional policies can be implemented by the Central Bank:

1. In order to reduce $M1$ (see equation (1)), legal reserve ratios r can be increased. In this case, banks have to decrease the amount of loans they provide, except for low-carbon projects that deliver carbon certificates. This solution increases borrowing rates for regular projects. Regular project investors consequently pay for low-carbon projects through more expensive loans.
2. Because most Central Banks follow an inflation-targeting strategy (Bernanke and Mishkin, 1997), the introduction of carbon certificates would induce them to anticipate an increase in inflation¹³ and to increase nominal interest rates (increased $\bar{r}$). In this case, the cost of capital is higher for all projects, except for low-carbon projects that are granted concessional loans in exchange for carbon certificates. Hence, as above, regular investors pay for low-carbon projects.

Both cases amount to the creation of differentiated interest rates, with low-carbon projects benefiting from lower capital costs than regular projects. In reality, the additional cost

of low-carbon projects is likely to be financed by a combination of increased economic activity, accelerated inflation, and higher capital cost for regular projects.

3. Conclusions

Eventually, this framework would have a redistributive impact among generations. Because it increases interest rates for carbon-emitting projects, it uses the wealth of future generations to pay for a hedge against a potentially dangerous climate change. In other words, reducing conventional investment in favour of consumption and mitigation is a way of shifting the costs of reducing emissions from the current generation to future generations, who will benefit from reduced climate change impacts.

An intragenerational distribution effect, which depends on whether inflation is increased or interest rates for regular projects are raised, might also be observed. However, the immediate, first-order effect would be much milder than that of a carbon tax, which would affect all economic actors rather than investors only.

Of course, the creation of carbon certificates is not as efficient as a carbon tax because it does not directly address consumption behaviour; it only internalizes part of the carbon externality through differentiated capital costs. In other words, the aggregate welfare cost of reaching the same target is higher than with a carbon tax. Also, efforts would be focused on capital-intensive solutions, which have no reason to be the most efficient and are impaired by rebound effects. However, given the amount of capital needed to shift investments towards clean technologies in energy, building, transport, and end-use equipment, it has the merit of developing these technologies and facilitating these investments, even in the absence of a carbon tax.

The objective of mitigating climate change interplays with the current financial and economic crisis. Carbon externalities (which are a transfer from future to current generations; e.g. Chichilnisky et al., 1995) should be a component of the broader reflection on the sustainability of public and private debts and the related pressure on current consumption levels. Also, any action on interest rates should be included in the design of short-term monetary policies – especially in the current European and American context – and Keynesian mechanisms need to be taken into account to assess the consequences of such a policy.

In a context of economic crisis, certificates could increase money creation (therefore increasing low-carbon investments) and, if Central Banks do not act to compensate for this additional money creation through 'sterilization' actions, this policy could also act as a 'green stimulus' policy (Zenghelis, 2011). In times of economic growth, the Central Bank can increase its rate to avoid undesired inflation. Choices concerning low-carbon investments can thus be separated from choices concerning monetary policy; creating carbon certificates does not constrain monetary policy, but shifts investments towards low-carbon projects.

Finally, it might be time to move away from what economists have presented as the 'optimal' policy, that is a uniform carbon tax, which appears politically compromised over the short term, at least in many countries. The scheme proposed here is a second-best solution that does not act on all possible levers. However, it seems easier to implement than a carbon tax, because its direct distributional impact on households and businesses appears milder. What justifies the introduction of this second-best option is more a policy failure than a market failure, namely the lack of political acceptability of a carbon tax. The introduction of carbon certificates as legal-reserve instruments and the creation of differentiated interest rates for low-carbon projects may appear as an interesting first step in the trajectory towards a low-carbon economy.

Acknowledgements

The authors wish to thank Céline Guivarch and three anonymous reviewers for insightful comments on a previous version of this paper. All remaining errors are the authors'. The views expressed in this article are the sole responsibility of the author. They do not necessarily reflect the views of the World Bank, its executive directors, nor the countries they represent.

Notes

1. This target might become more stringent because the EU 'Roadmap for moving to a low-carbon economy in 2050' has moved to an objective of at least 25% reduction in GHG emissions by 2020.
2. The proportional increase in investment needs differs depending on the sector under consideration; for

example, it is likely to be relatively larger in the energy sector.

3. By introducing a price for carbon, the profitability of 'regular' capital falls, because it now has to pay for the carbon emissions. Low-carbon capital becomes more profitable and investment funds can be obtained from the market more easily.
4. In fact, correcting one externality always increases welfare only if there is just one externality. In the presence of multiple externalities, correcting one of them can reduce welfare (Lipsey and Lancaster, 1956).
5. A carbon market with full auctioning of allowances, and covering all economic sectors, is also an unrealistic option in the short term.
6. This solution is Pareto-improving only if intragenerational distributional effects can be compensated through *ad hoc* measures or pre-existing safety nets.
7. Even though production Y is affected by ΔY , consumption C is maintained by reducing investment I by ΔY .
8. The welfare derived from consumption would be preserved, even though the structure of consumption would be modified by changes in the relative prices.
9. The increase in low-carbon investments (and their extra costs relative to regular projects) would be compensated by a decrease in regular investments.
10. Note that this article does not touch on the choice of mitigation targets, but deals only with ways of reaching a predetermined target. These two aspects are related, for example through the discount rate, which influences the weight of future climate change impacts on current decisions.
11. Application at the international level combined with a reform of the international monetary system is discussed in Hourcade et al. (2011).
12. However, accelerated inflation will have a negative effect on the exchange rate. This effect can in turn decrease the attractiveness for investors, with consequences for employment and growth, as well as carbon leakage. These international aspects require further research to assess their significance.
13. Note that if inflation increase is a surprise, the Central Banks would react afterwards but increase interest rates anyway, in order to lower inflation to its target.

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